

- 1. Prerequisites** A knowledge of the specifications for S1 and S2, and their prerequisites and associated formulae, is assumed and may be tested.
- 2. Examination**
- First assessment: June 2020.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 75 marks.
 - Students must answer all questions.
 - Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
 - The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
- 3. Notation and formulae** Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.
- Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.
- $aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$ where X and Y are independent and $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$.
-

S3.3 Unit content

What students need to learn:		Guidance
1. Combinations of random variables		
1.1	Distribution of linear combinations of independent Normal random variables.	If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ independently, then $aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$. No proofs required.